

A photograph of a modern university building with large glass windows and a white facade, situated behind a field of tall grass and yellow wildflowers under a blue sky with scattered clouds. The text is overlaid on a semi-transparent grey box.

5CM512

# Communication and Security Protocols

**Week 6:**

**Packets: IP Security**

# Basics of IPSec

IPSec is a protocol suite to authenticate and encrypt the packets being exchanged between two points.

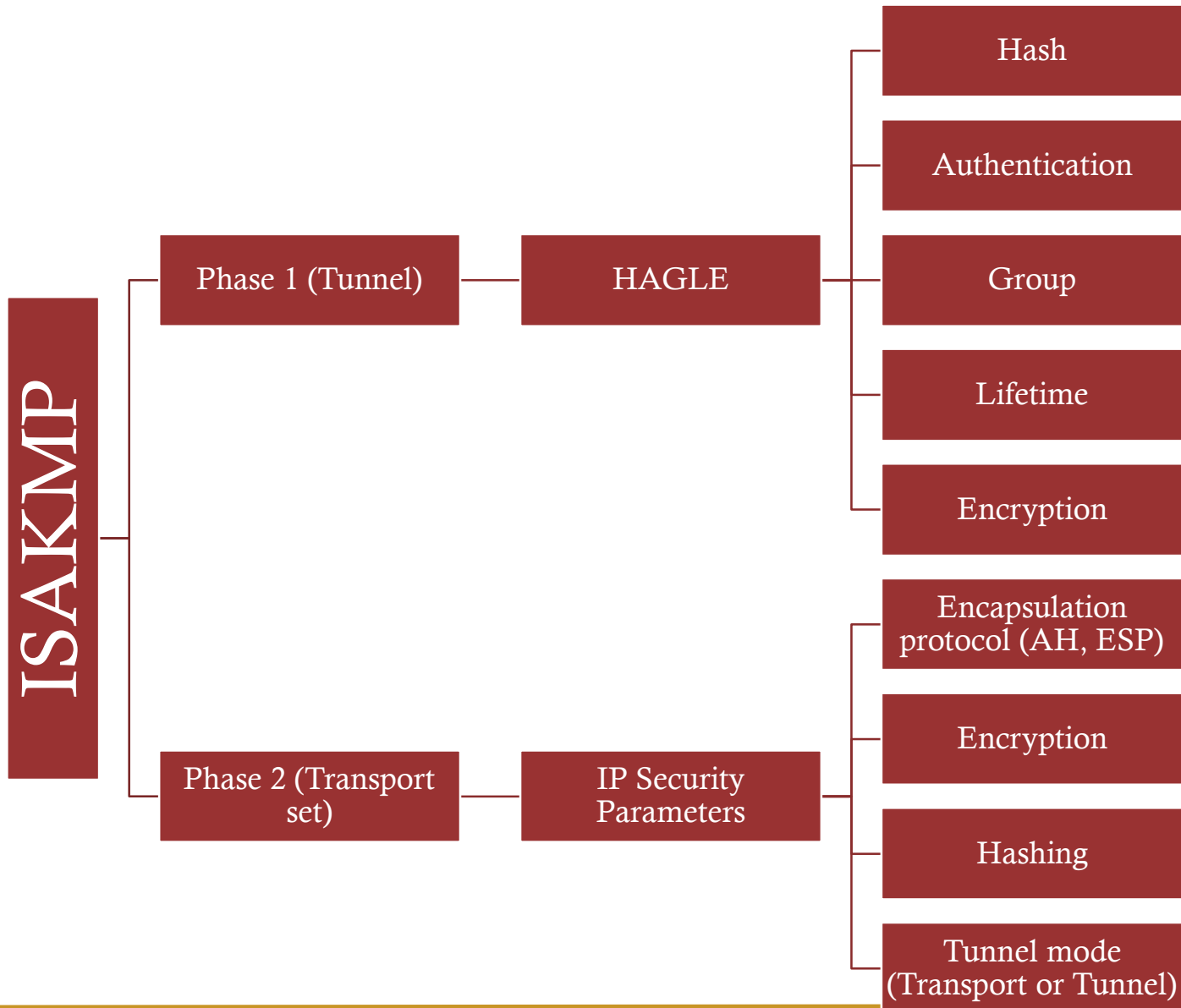
- Integrity: It indicates that the received message is same message that was sent [MD5, SHA]
- Authentication: It refers to verifying the identity of a network entity like user/device [PSK, RSA]
- Confidentiality: It is used to hide information [DES, AES]

# IPSec

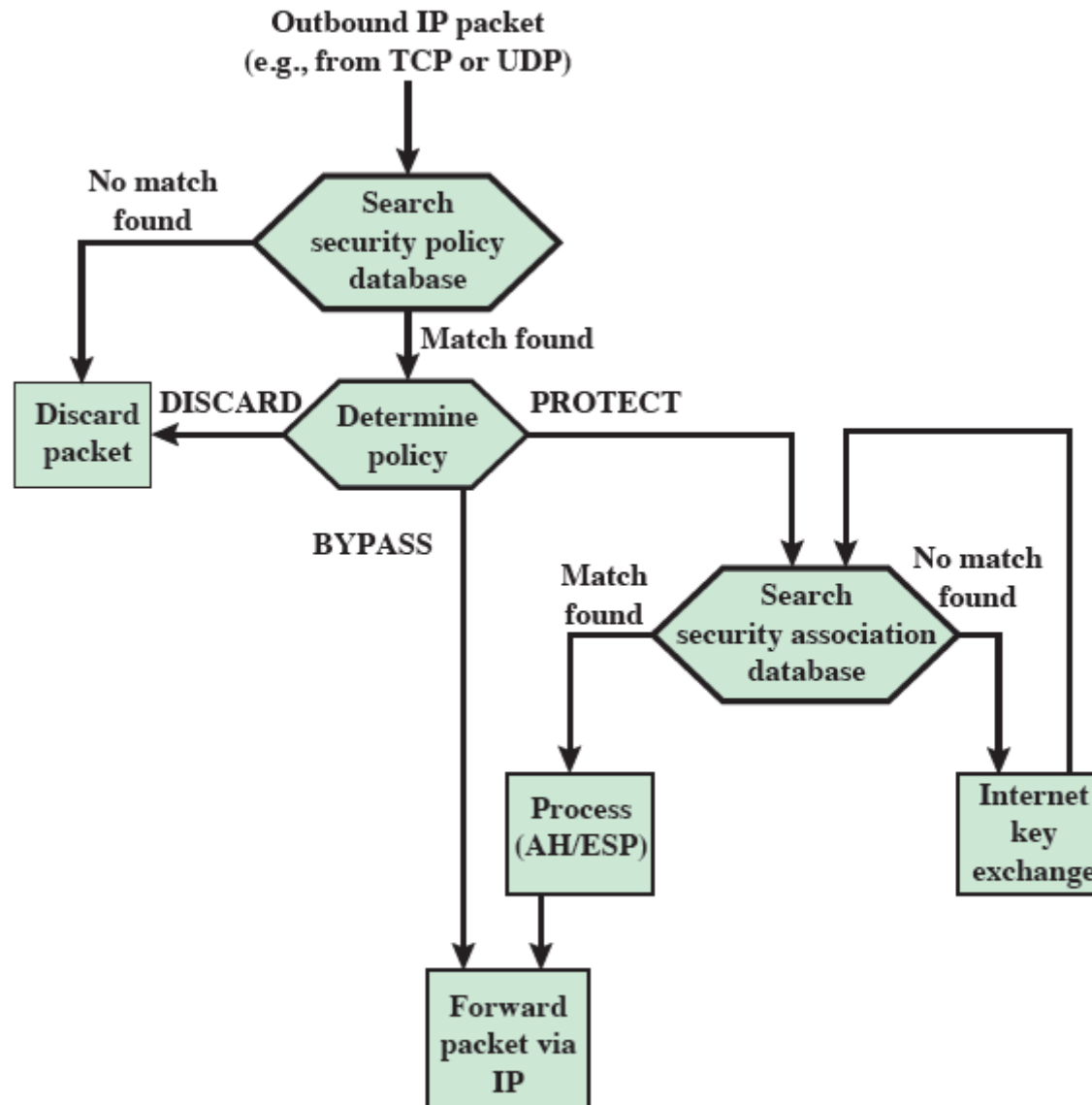
Security  
Protocols  
(AH, ESP)

Models  
(Transport,  
Tunnel)

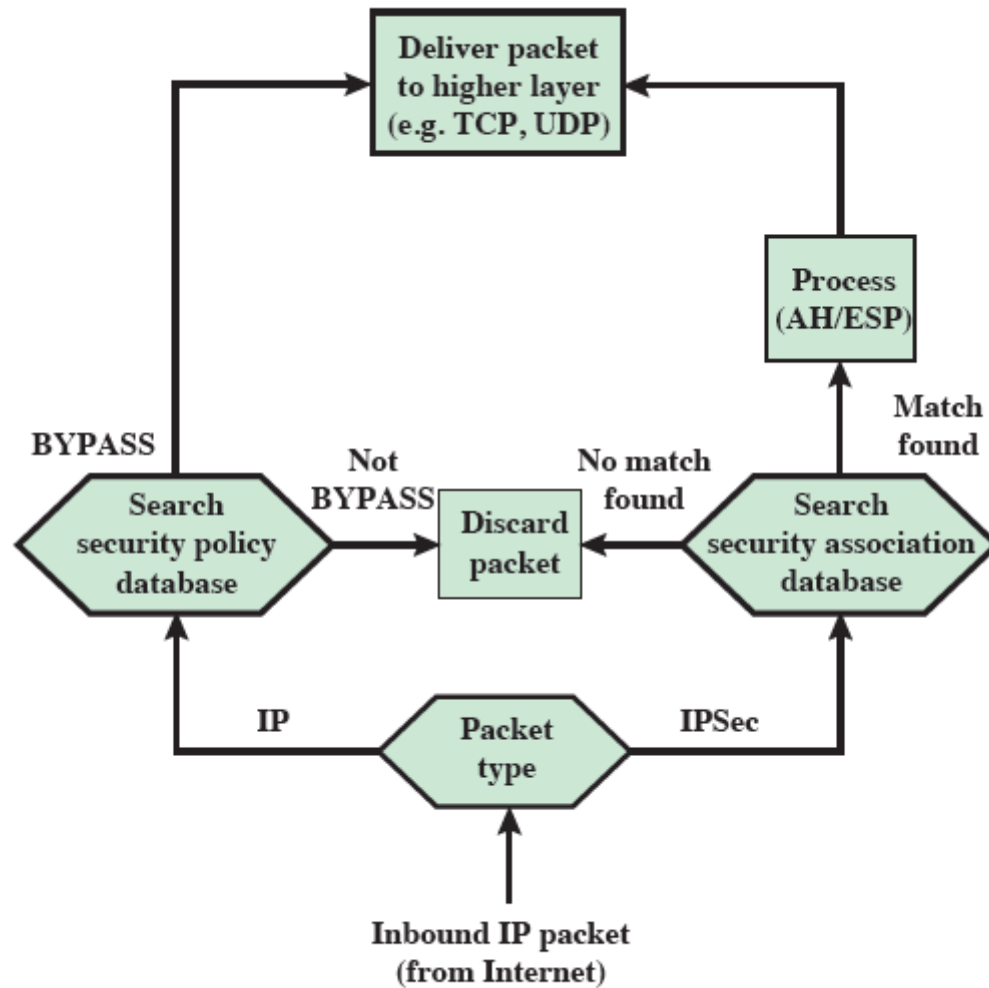
Key  
management  
(SA)



# Processing Model for Outbound Packets



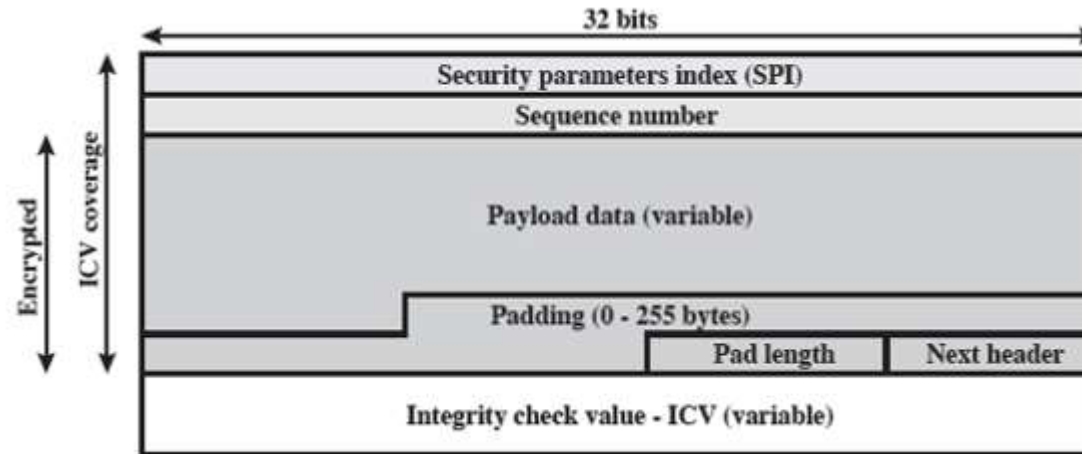
# Processing Model for Inbound Packets



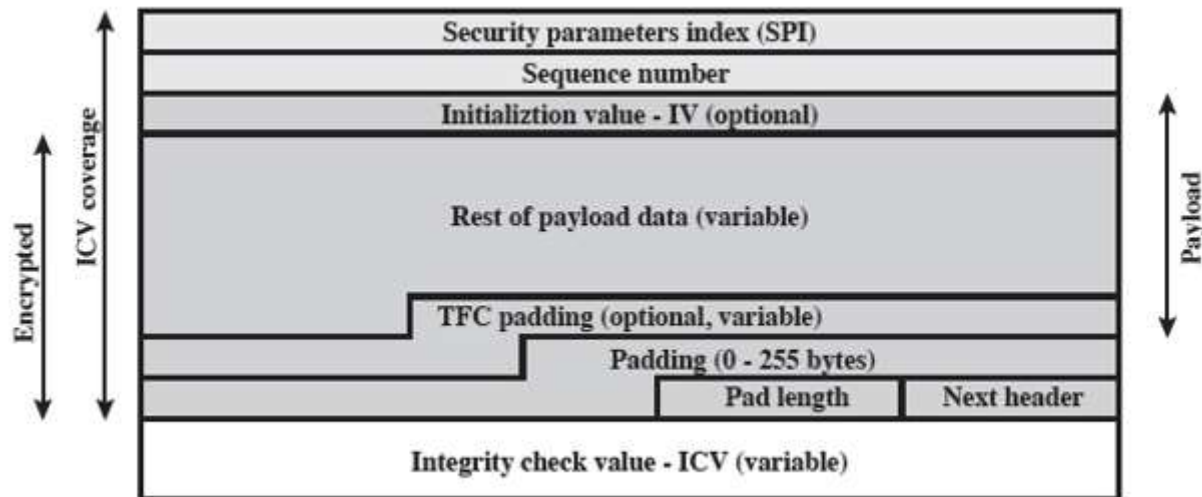


# ESP Packet Format

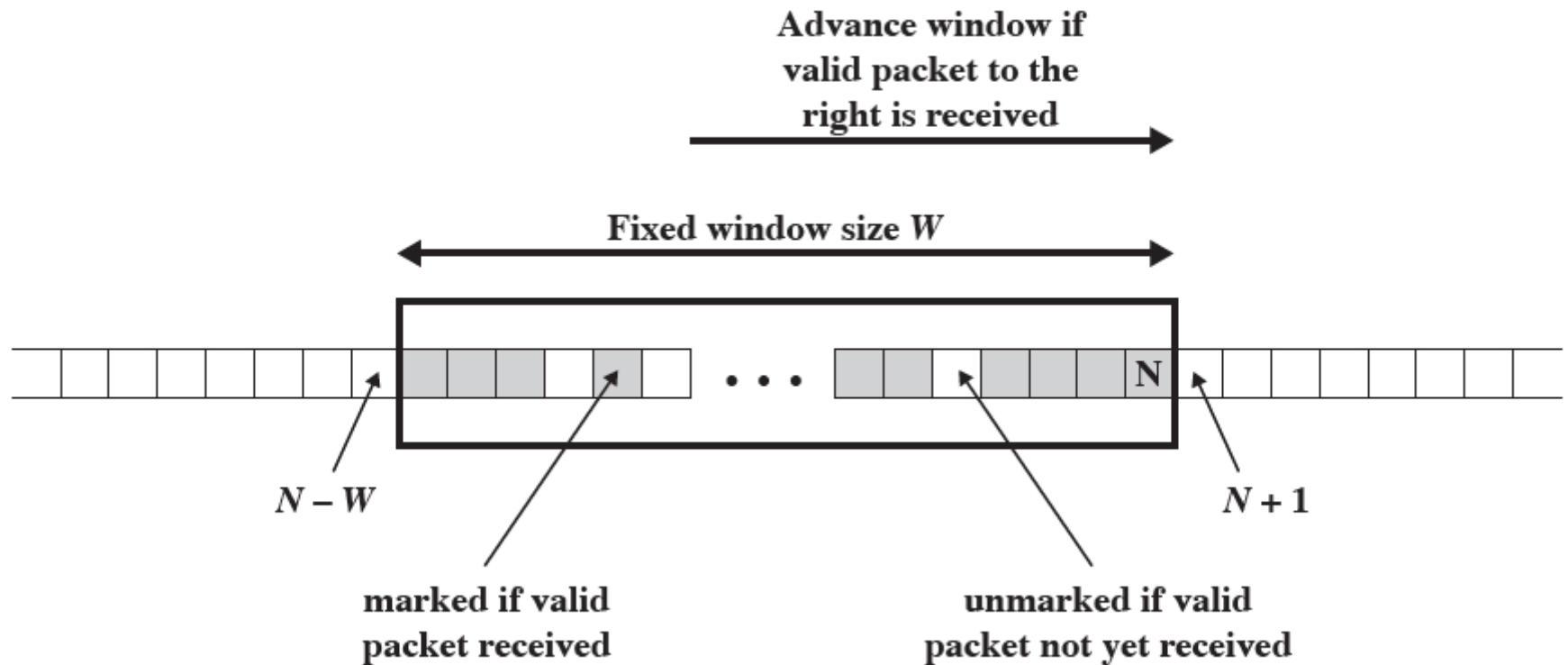
Top Level format



Substructure of Payload Data



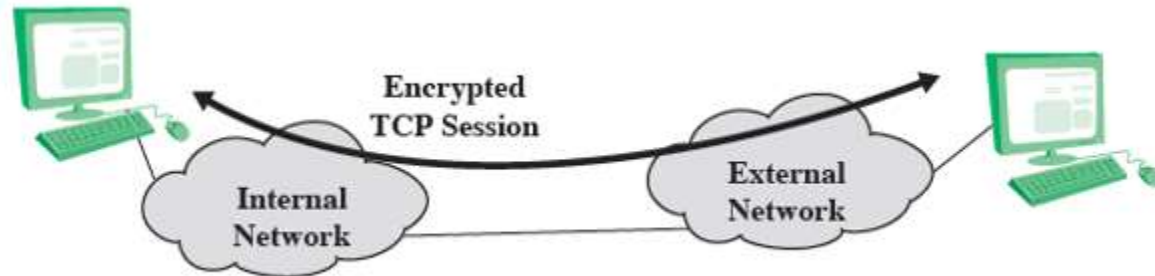
# Anti-Replay Mechanism



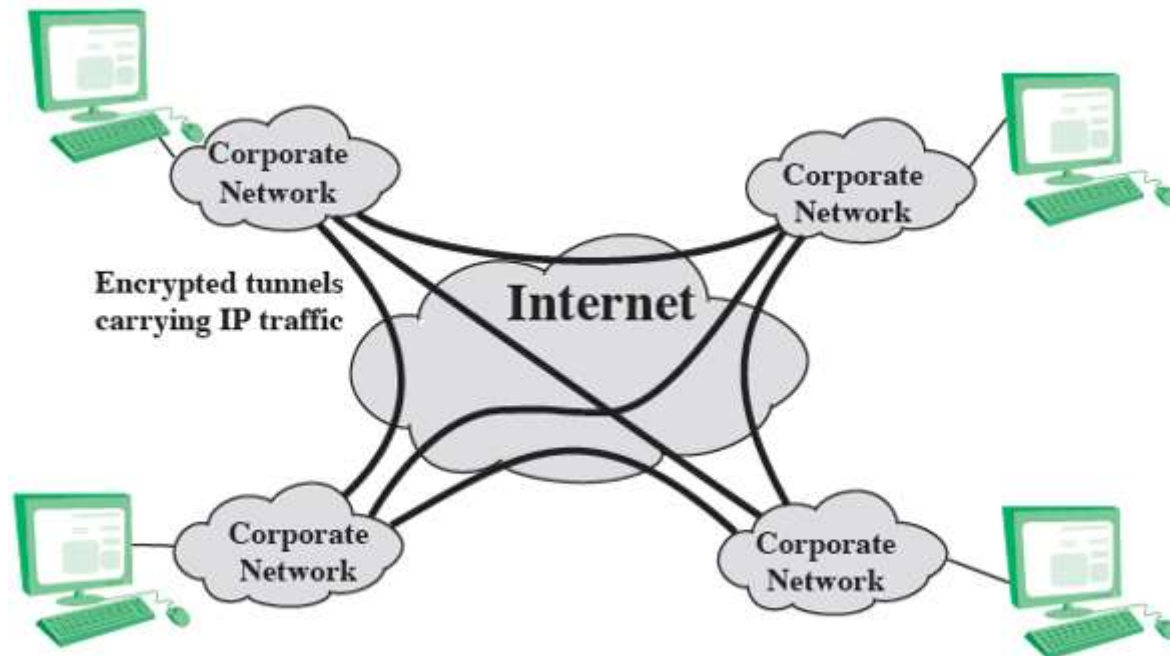


# Transport-Mode vs. Tunnel-Mode Encryption

Transport level security

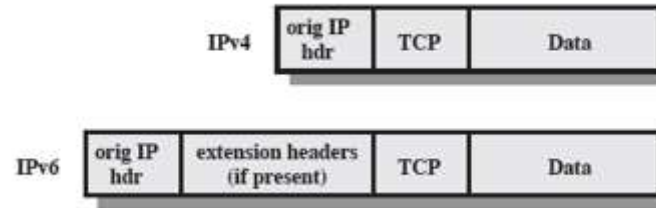


Virtual Private Network (VPN) via Tunnel Mode

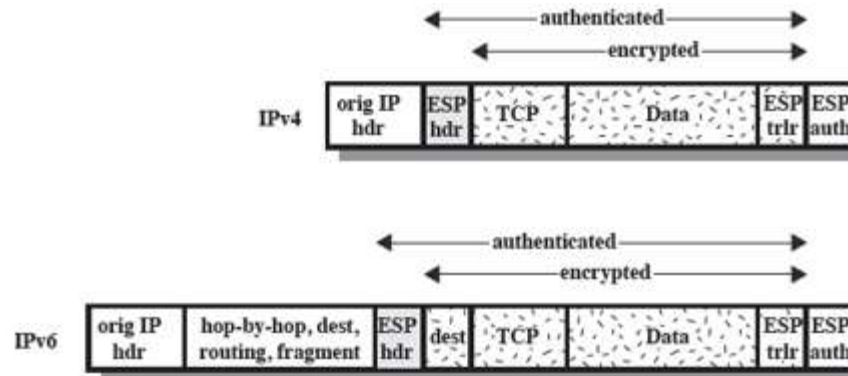


# Scope of ESP Encryption and Authentication

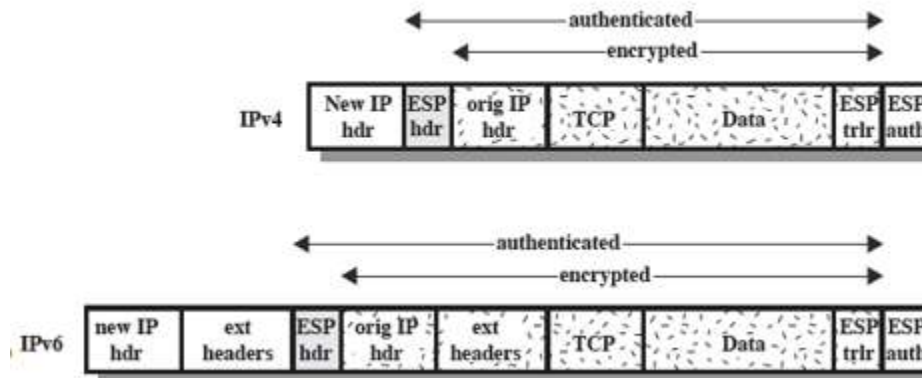
Before Applying ESP



Transport mode

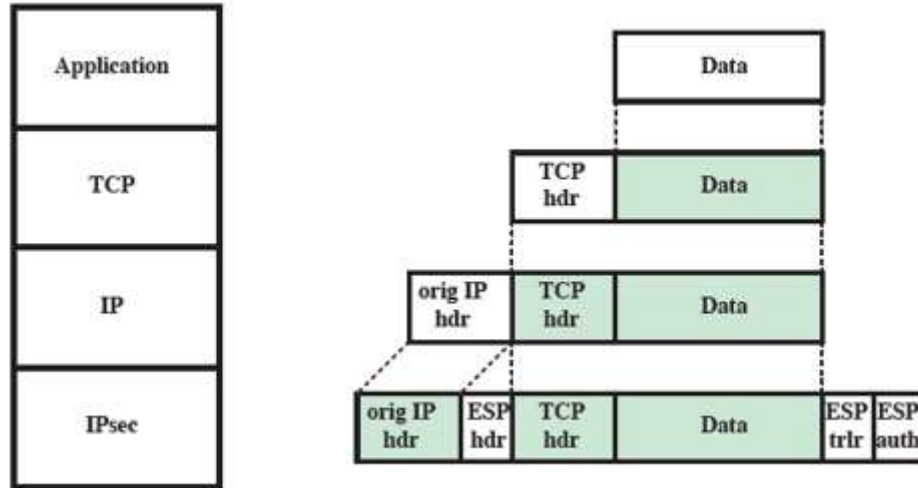


Tunnel mode

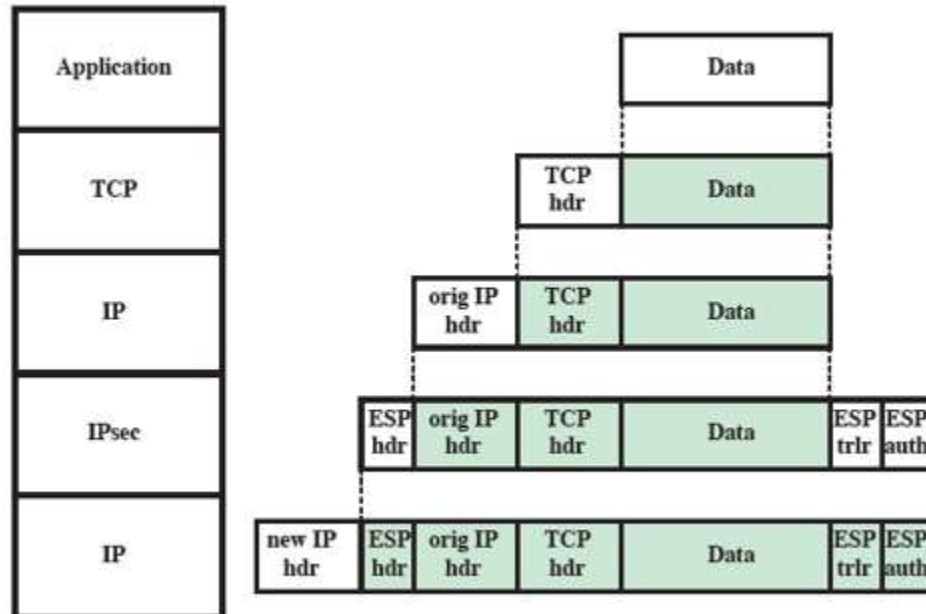


# Protocol Operation for ESP

Transport mode



Tunnel mode



# Combining Security Associations

- An individual SA can implement either the AH or ESP protocol but not both
- *Security association bundle*
  - Refers to a sequence of SAs through which traffic must be processed to provide a desired set of IPsec services
  - The SAs in a bundle may terminate at different endpoints or at the same endpoint
- May be combined into bundles in two ways:

## Transport adjacency

- Refers to applying more than one security protocol to the same IP packet without invoking tunneling
- This approach allows for only one level of combination

## Iterated tunneling

- Refers to the application of multiple layers of security protocols effected through IP tunneling
- This approach allows for multiple levels of nesting

# ESP with Authentication Option

- In this approach, the first user applies ESP to the data to be protected and then appends the authentication data field

## Transport mode ESP

- Authentication and encryption apply to the IP payload delivered to the host, but the IP header is not protected

## Tunnel mode ESP

- Authentication applies to the entire IP packet delivered to the outer IP destination address and authentication is performed at that destination
- The entire inner IP packet is protected by the privacy mechanism for delivery to the inner IP destination

- For both cases authentication applies to the ciphertext rather than the plaintext

# Transport Adjacency

- Another way to apply authentication after encryption is to use two bundled transport SAs, with the inner being an ESP SA and the outer being an AH SA
  - In this case ESP is used without its authentication option
  - Encryption is applied to the IP payload
  - AH is then applied in transport mode
  - Advantage of this approach is that the authentication covers more fields
  - Disadvantage is the overhead of two SAs versus one SA

# Internet Key Exchange

- The key management portion of IPsec involves the determination and distribution of secret keys
- A typical requirement is four keys for communication between two applications
  - Transmit and receive pairs for both integrity and confidentiality

The IPsec Architecture document mandates support for two types of key management:

- A system administrator manually configures each system with its own keys and with the keys of other communicating systems
- This is practical for small, relatively static environments

Manual

Automated

- Enables the on-demand creation of keys for SAs and facilitates the use of keys in a large distributed system with an evolving configuration



# ISAKMP/Oakley

- The default automated key management protocol of IPsec
- Consists of:
  - Oakley Key Determination Protocol
    - A key exchange protocol based on the Diffie-Hellman algorithm but providing added security
    - Generic in that it does not dictate specific formats
  - Internet Security Association and Key Management Protocol (ISAKMP)
    - Provides a framework for Internet key management and provides the specific protocol support, including formats, for negotiation of security attributes
    - Consists of a set of message types that enable the use of a variety of key exchange algorithms

# Features of IKE Key Determination

- Algorithm is characterized by five important features:

1.

- It employs a mechanism known as cookies to thwart clogging attacks

2.

- It enables the two parties to negotiate a group; this, in essence, specifies the global parameters of the Diffie-Hellman key exchange

3.

- It uses nonces to ensure against replay attacks

4.

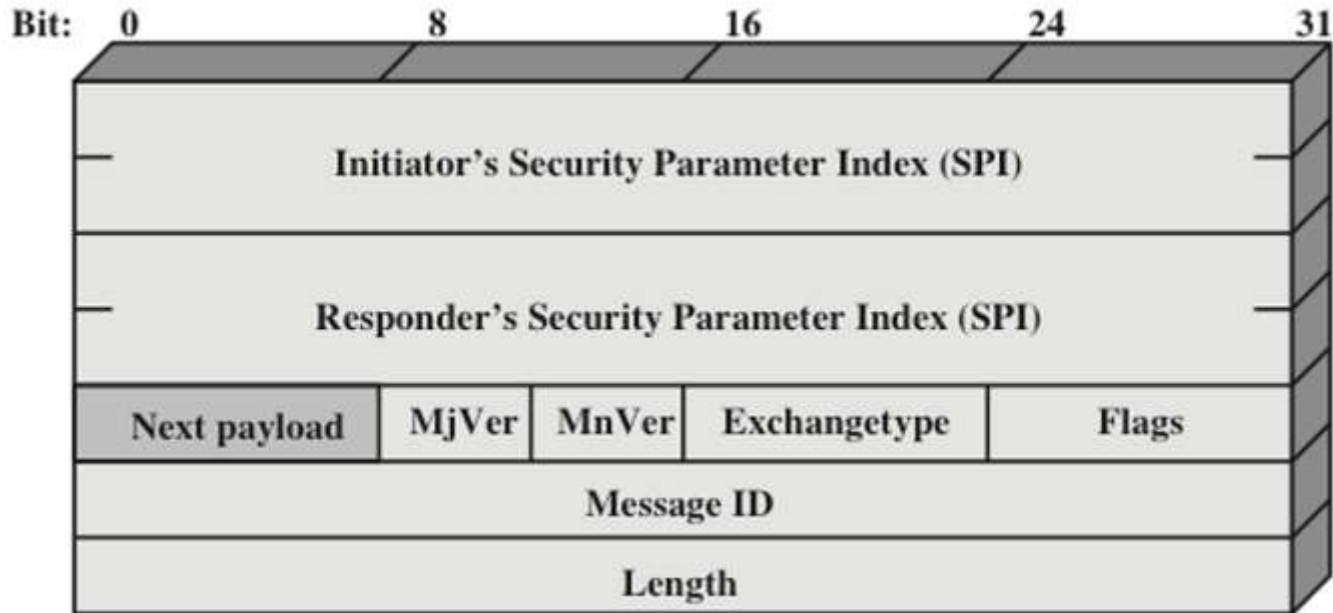
- It enables the exchange of Diffie-Hellman public key values

5.

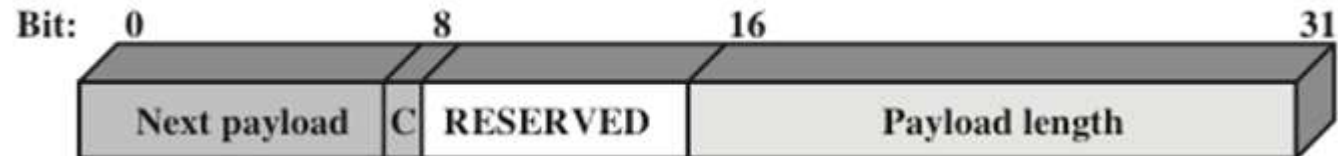
- It authenticates the Diffie-Hellman exchange to thwart man-in-the-middle-attacks

# IKE Formats

IKE Header



Generic Payload Header



# IKE Payload Types

| Type                               | Parameters                                                         |
|------------------------------------|--------------------------------------------------------------------|
| Security Association               | Proposals                                                          |
| Key Exchange                       | DH Group #, Key Exchange Data                                      |
| Identification                     | ID Type, ID Data                                                   |
| Certificate                        | Cert Encoding, Certificate Data                                    |
| Certificate Request                | Cert Encoding, Certification Authority                             |
| Authentication                     | Auth Method, Authentication Data                                   |
| Nonce                              | Nonce Data                                                         |
| Notify                             | Protocol-ID, SPI Size, Notify Message Type, SPI, Notification Data |
| Delete                             | Protocol-ID, SPI Size, # of SPIs, SPI (one or more)                |
| Vendor ID                          | Vendor ID                                                          |
| Traffic Selector                   | Number of TSs, Traffic Selectors                                   |
| Encrypted                          | IV, Encrypted IKE payloads, Padding, Pad Length, ICV               |
| Configuration                      | CFG Type, Configuration Attributes                                 |
| Extensible Authentication Protocol | EAP Message                                                        |

## Cryptographic Suites for IPsec

|                | <b>VPN-A</b>  | <b>VPN-B</b>          |
|----------------|---------------|-----------------------|
| ESP encryption | 3DES-CBC      | AES-CBC (128-bit key) |
| ESP integrity  | HMAC-SHA1-96  | AES-XCBC-MAC-96       |
| IKE encryption | 3DES-CBC      | AES-CBC (128-bit key) |
| IKE PRF        | HMAC-SHA1     | AES-XCBC-PRF-128      |
| IKE Integrity  | HMAC-SHA1-96  | AES-XCBC-MAC-96       |
| IKE DH group   | 1024-bit MODP | 2048-bit MODP         |

(a) Virtual private networks (RFC 4308)

|                              | <b>GCM-128</b>           | <b>GCM-256</b>           | <b>GMAC-128</b>            | <b>GMAC-256</b>           |
|------------------------------|--------------------------|--------------------------|----------------------------|---------------------------|
| ESP encryption/<br>Integrity | AES-GCM<br>(128-bit key) | AES-GCM<br>(256-bit key) | Null                       | Null                      |
| ESP integrity                | Null                     | Null                     | AES-GMAC (128-<br>bit key) | AES-GMAC<br>(256-bit key) |
| IKE encryption               | AES-CBC<br>(128-bit key) | AES-CBC<br>(256-bit key) | AES-CBC<br>(128-bit key)   | AES-CBC<br>(256-bit key)  |
| IKE PRF                      | HMAC-SHA-256             | HMAC-SHA-384             | HMAC-SHA-256               | HMAC-SHA-384              |
| IKE Integrity                | HMAC-SHA-<br>256-128     | HMAC-SHA-<br>384-192     | HMAC-SHA-<br>256-128       | HMAC-SHA-<br>384-192      |
| IKE DH group                 | 256-bit random<br>ECP    | 384-bit random<br>ECP    | 256-bit random<br>ECP      | 384-bit random<br>ECP     |

(b) NSA Suite B (RFC 6379)

# Any QUESTIONS?

